

## **8)Network Virtualization – OpenVSwitch and Linux bridge**

### **Aim:**

To perform network virtualization in Linux using OpenVswitch and Linux Bridge.

### OpenVswitch

OpenVSwitch (OVS) is an open-source, production-quality, multi-layer software switch designed for virtualized environments, particularly in Linux-based hypervisors. It provides virtual network connectivity to virtual machines (VMs) and serves as a foundational component in modern data centers and software-defined networks (SDNs). OVS enables network automation through programmatic extensions like the OpenFlow protocol and the OVSDB management protocol, while also supporting standard network management interfaces such as NetFlow, sFlow, and LACP

### Linux bridge

A Linux bridge is a software-based network device that functions like a physical network switch, allowing multiple network interfaces (physical or virtual) to be connected and operate as a single logical network segment. It works at the data link layer (Layer 2), forwarding network traffic based on MAC addresses rather than IP addresses. Linux bridges are crucial for virtual networking, enabling virtual machines and containers to communicate with each other and with the physical network, and are often used to create isolated networks or connect different network segments

## PROCEDURE:

### Linux Bridge:

1. Check the current network configuration.

```
tce@tce-VirtualBox:~/praveen111$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255
    inet6 fe80::3afe:57b2:7efb:319b prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:19:cc:34 txqueuelen 1000 (Ethernet)
    RX packets 672 bytes 772620 (772.6 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 384 bytes 50076 (50.0 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

2. Install and configure necessary Bridge Utilities

- a. Install bridge-utils

```
tce@tce-VirtualBox:~/praveen111$ sudo apt install bridge-utils
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following NEW packages will be installed:
  bridge-utils
```

- b. Load loads the bridge kernel module, which is required for bridge functionality.

```
tce@tce-VirtualBox:~/praveen111$ sudo modprobe bridge
```

- c. Loads the br\_netfilter kernel module, which enables netfilter (firewall) rules to be applied to traffic passing through the bridge.

```
tce@tce-VirtualBox:~/praveen111$ sudo modprobe br_netfilter
```

3. Edit the netplan configuration file and specify the configuration.

```

network:
  version:2
  renderer: networkd
  ethernets:
    eno2:
      dhcp4: no
  bridges:
    br-cloud:
      interfaces: [eno2]
      addresses: [10.0.2.15/24]
      gateway4: 10.0.2.255
      nameservers:
        addresses: [8.8.8.8,1.1.1.1]

```

4. Apply the new netplan configuration and restart the network.

```

tce@tce-VirtualBox:~/praveen111$ sudo nano /etc/netplan/01-netcfg.yaml
tce@tce-VirtualBox:~/praveen111$ sudo systemctl restart networking

```

5. Verify the network interfaces and bridges

```

tce@tce-VirtualBox:~/praveen111$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:19:cc:34 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
        valid_lft 85630sec preferred_lft 85630sec
    inet6 fe80::3afe:57b2:7efb:319b/64 scope link noprefixroute
        valid_lft forever preferred_lft forever

```

6. Verify the routing table

```

tce@tce-VirtualBox:~/praveen111$ route -n
Kernel IP routing table
Destination      Gateway         Genmask         Flags Metric Ref    Use Iface
0.0.0.0          10.0.2.2        0.0.0.0         UG    20425  0      0 br-clou
d
10.0.2.0         0.0.0.0         255.255.255.0   U     425    0      0 br-clou
d
169.254.0.0      0.0.0.0         255.255.0.0     U     1000   0      0 br-clou
d

```

```

tce@tce-VirtualBox:~$ brctl show
bridge name      bridge id                STP enabled  interfaces
br-cloud         8000.000000000000        no
tce@tce-VirtualBox:~$

```

Alternatively, you can use the following commands to configure the linux bridge

**Open Virtual Switch (OpenVSwitch/OVS):**

1. Enter as root user.

```
tce@tce-VirtualBox:~/praveen111$ sudo -s
```

2. Create 2 namespace (in my case VRF1 and VRF2)

```
root@tce-VirtualBox:~/praveen111# ip netns add VRF1
root@tce-VirtualBox:~/praveen111# ip netns add VRF2
root@tce-VirtualBox:~/praveen111# ip netns list
VRF2
VRF1
```

3. Create 4 Virtual ethernet port to attach with bridge and namespaces.

```
root@tce-VirtualBox:~/praveen111# ip link add veth1 type veth peer name veth2
root@tce-VirtualBox:~/praveen111# ip link add veth3 type veth peer name veth4
```

Similarly add veth3 and veth4.

4. If the configuration is done to create the topology shown in the image, the links will be as

```
5: veth2@veth1: <BROADCAST,MULTICAST,M-DOWN> mtu 1500 qdisc noop state DOWN gro
up default qlen 1000
    link/ether 16:79:88:69:e0:db brd ff:ff:ff:ff:ff:ff
6: veth1@veth2: <BROADCAST,MULTICAST,M-DOWN> mtu 1500 qdisc noop state DOWN gro
up default qlen 1000
    link/ether ba:78:a7:b0:63:d7 brd ff:ff:ff:ff:ff:ff
7: veth4@veth3: <BROADCAST,MULTICAST,M-DOWN> mtu 1500 qdisc noop state DOWN gro
up default qlen 1000
    link/ether 42:23:d1:70:a1:cf brd ff:ff:ff:ff:ff:ff
8: veth3@veth4: <BROADCAST,MULTICAST,M-DOWN> mtu 1500 qdisc noop state DOWN gro
up default qlen 1000
    link/ether a6:6b:68:42:5b:04 brd ff:ff:ff:ff:ff:ff
root@tce-VirtualBox:~/praveen111#
```

5. Assign each virtual ethernet port to the particular namespace

```
root@tce-VirtualBox:~/praveen111# ip link set veth1 netns VRF1
root@tce-VirtualBox:~/praveen111# ip link set veth3 netns VRF2
root@tce-VirtualBox:~/praveen111# ip link
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT
    T group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel master br-
cloud state UP mode DEFAULT group default qlen 1000
    link/ether 08:00:27:19:cc:34 brd ff:ff:ff:ff:ff:ff
4: br-cloud: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP
    mode DEFAULT group default qlen 1000
    link/ether 08:00:27:19:cc:34 brd ff:ff:ff:ff:ff:ff
5: veth2@if6: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN mode DEFAULT
    group default qlen 1000
    link/ether 16:79:88:69:e0:db brd ff:ff:ff:ff:ff:ff link-netnsid 0
7: veth4@if8: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN mode DEFAULT
    group default qlen 1000
    link/ether 42:23:d1:70:a1:cf brd ff:ff:ff:ff:ff:ff link-netnsid 1
root@tce-VirtualBox:~/praveen111#
```

```

root@tce-VirtualBox:~/praveen111# ip netns exec VRF1 ip link
1: lo: <LOOPBACK> mtu 65536 qdisc noop state DOWN mode DEFAULT group default ql
en 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
6: veth1@if5: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN mode DEFAULT
group default qlen 1000
    link/ether ba:78:a7:b0:63:d7 brd ff:ff:ff:ff:ff:ff link-netnsid 0
root@tce-VirtualBox:~/praveen111#

```

You can see that veth1 port is attached to the VRF1 namespace.

## 6. Assign IP addresses to the virtual ethernet ports

```

root@tce-VirtualBox:~/praveen111# ip netns exec VRF1 ifconfig veth1 10.10.10.1/
24 up
root@tce-VirtualBox:~/praveen111# ip netns exec VRF2 ifconfig veth3 10.10.10.1/
24 up
root@tce-VirtualBox:~/praveen111# ip netns exec VRF2 ifconfig
veth3: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500
    inet 10.10.10.1 netmask 255.255.255.0 broadcast 10.10.10.255
    ether a6:6b:68:42:5b:04 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

root@tce-VirtualBox:~/praveen111# ip netns exec VRF1 ifconfig
veth1: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500
    inet 10.10.10.1 netmask 255.255.255.0 broadcast 10.10.10.255
    ether ba:78:a7:b0:63:d7 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

root@tce-VirtualBox:~/praveen111#

```

## 7. Check the configuration using the default routing table

```

root@tce-VirtualBox:~/praveen111# ip netns exec VRF2 route -n
Kernel IP routing table
Destination Gateway Genmask Flags Metric Ref Use Iface
10.10.10.0 0.0.0.0 255.255.255.0 U 0 0 0 veth3
root@tce-VirtualBox:~/praveen111# route -n
Kernel IP routing table
Destination Gateway Genmask Flags Metric Ref Use Iface
0.0.0.0 10.0.2.2 0.0.0.0 UG 425 0 0 br-clou
d
10.0.2.0 0.0.0.0 255.255.255.0 U 425 0 0 br-clou
d
169.254.0.0 0.0.0.0 255.255.0.0 U 1000 0 0 br-clou
d
root@tce-VirtualBox:~/praveen111# ip netns exec VRF1 route -n
Kernel IP routing table
Destination Gateway Genmask Flags Metric Ref Use Iface
10.10.10.0 0.0.0.0 255.255.255.0 U 0 0 0 veth1
root@tce-VirtualBox:~/praveen111#

```

## 8. Install openVswitch and its dependencies

```

root@tce-VirtualBox:~/praveen111# sudo apt install openvswitch-switch openvswit
ch-common -y
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  libpython-stdlib libpython2.7 libpython2.7-minimal libpython2.7-stdlib
  python python-minimal python-six python2.7 python2.7-minimal
Suggested packages:
  ethtool openvswitch-doc python-doc python-tk python2.7-doc binfmt-support
The following NEW packages will be installed:
  libpython-stdlib openvswitch-common openvswitch-switch python
  python-minimal python-six python2.7 python2.7-minimal
The following packages will be upgraded:

```

9. Check the status of the openvswitch service, if off turn it on.

```

root@tce-VirtualBox:~/praveen111# ovs-vsctl --version
ovs-vsctl (Open vSwitch) 2.9.8
DB Schema 7.15.1
root@tce-VirtualBox:~/praveen111# sudo systemctl status openvswitch-switch
Unit openvswitch-switch.service could not be found.
root@tce-VirtualBox:~/praveen111# sudo systemctl status openvswitch-switch
● openvswitch-switch.service - Open vSwitch
   Loaded: loaded (/lib/systemd/system/openvswitch-switch.service; enabled; ven
   Active: active (exited) since Mon 2026-09-07 16:12:11 IST; 1min 32s ago
   Main PID: 5787 (code=exited, status=0/SUCCESS)
     Tasks: 0 (limit: 2326)
    CGroup: /system.slice/openvswitch-switch.service

Sep 07 16:12:11 tce-VirtualBox systemd[1]: Starting Open vSwitch...
Sep 07 16:12:11 tce-VirtualBox systemd[1]: Started Open vSwitch.
root@tce-VirtualBox:~/praveen111#

```

10. Add the switch using ovscommandline tools and check the configuration

```

root@tce-VirtualBox:~/praveen111# ovs-vsctl add-br vSwitch1
root@tce-VirtualBox:~/praveen111# ovs-vsctl show
775ca24e-cbe1-426a-a377-6fa1706bf073
    Bridge "vSwitch1"
        Port "vSwitch1"
            Interface "vSwitch1"
                type: internal
    ovs_version: "2.9.8"
root@tce-VirtualBox:~/praveen111# ovs-vsctl add-port vSwitch1 veth2
root@tce-VirtualBox:~/praveen111# ovs-vsctl add-port vSwitch1 veth4
root@tce-VirtualBox:~/praveen111# ovs-vsctl show
775ca24e-cbe1-426a-a377-6fa1706bf073
    Bridge "vSwitch1"
        Port "veth4"
            Interface "veth4"
        Port "vSwitch1"
            Interface "vSwitch1"
                type: internal
        Port "veth2"
            Interface "veth2"
    ovs_version: "2.9.8"
root@tce-VirtualBox:~/praveen111#

```

11. Turn the virtual ethernet ports up and configure IP for switch check their status

```

root@tce-VirtualBox:~/praveen111# ifconfig veth2 up
root@tce-VirtualBox:~/praveen111# ifconfig veth4 up
root@tce-VirtualBox:~/praveen111# ifconfig vSwitch1 10.10.10.3/24 up
root@tce-VirtualBox:~/praveen111# ip link

```

```

vSwitch1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
  inet 10.10.10.3 netmask 255.255.255.0 broadcast 10.10.10.255
  inet6 fe80::40c7:15ff:fe75:4b49 prefixlen 64 scopeid 0x20<link>
  ether 42:c7:15:75:4b:49 txqueuelen 1000 (Ethernet)
  RX packets 2 bytes 112 (112.0 B)
  RX errors 0 dropped 16 overruns 0 frame 0
  TX packets 53 bytes 6742 (6.7 KB)
  TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

veth2: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
  inet6 fe80::1479:88ff:fe69:e0db prefixlen 64 scopeid 0x20<link>
  ether 16:79:88:69:e0:db txqueuelen 1000 (Ethernet)
  RX packets 9 bytes 726 (726.0 B)
  RX errors 0 dropped 0 overruns 0 frame 0
  TX packets 85 bytes 10320 (10.3 KB)
  TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

veth4: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
  inet6 fe80::4023:d1ff:fe70:a1cf prefixlen 64 scopeid 0x20<link>
  ether 42:23:d1:70:a1:cf txqueuelen 1000 (Ethernet)
  RX packets 9 bytes 726 (726.0 B)
  RX errors 0 dropped 0 overruns 0 frame 0
  TX packets 79 bytes 9804 (9.8 KB)
  TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

root@tce-VirtualBox:~/praveen111#

```

12. Ping the namespace at the other end, if the configuration is done correct we can ping the Virtual namespace 2 from virtual namespace 1 via the switch (ovs bridge)

```

root@tce-VirtualBox:~/praveen111# sudo ip netns exec VRF1 ping -c 4 10.10.10.3
PING 10.10.10.3 (10.10.10.3) 56(84) bytes of data:
64 bytes from 10.10.10.3: icmp_seq=1 ttl=64 time=0.299 ms
64 bytes from 10.10.10.3: icmp_seq=2 ttl=64 time=0.056 ms
64 bytes from 10.10.10.3: icmp_seq=3 ttl=64 time=0.092 ms
^C
--- 10.10.10.3 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2041ms
rtt min/avg/max/mdev = 0.056/0.149/0.299/0.107 ms

```

13. Optionally you can turn on the spanning tree protocol (to prevent broadcast storms and network loops)

```

root@tce-VirtualBox:~/praveen111# ovs-vsctl set bridge vSwitch1 stp_enable=true
root@tce-VirtualBox:~/praveen111# ovsdb-client dump
AutoAttach table
 _uuid mappings system_description system_name
-----
Bridge table
 _uuid auto_attach controller datapath_id
datapath_type datapath_version external_ids fail_mode flood_vlans flow_tables i
pfx mcast_snooping_enable mirrors name netflow other_config ports
protocols rstp_enable rstp_status sflow status

stp_enable
-----
7515c743-3f47-494b-8f83-196a3795976d [] [] "000042c715754b49"
"" "<unknown>" {} [] [] {} [
] false [] "vSwitch1" [] {} [2ecafb08-f3
8f-44d7-9128-5844170a90e3, 38b6241c-a987-43f3-b33e-e4af9f68e482, 9d4292c9-03bd-
44f7-ab49-08d353603d4a] [] false {} [] {stp bridge id=

```

14. Finally check the MAC address attached with the switch and the actual mac address of the virtual namespace device, if they match everything is done right.

```
root@tce-VirtualBox:~/praveen111# ovs-appctl fdb/show vSwitch1
port  VLAN  MAC                Age
  2      0  a6:6b:68:42:5b:04   195
  1      0  ba:78:a7:b0:63:d7   63
LOCAL   0  42:c7:15:75:4b:49   63
root@tce-VirtualBox:~/praveen111# ip netns exec VRF2 ifconfig
veth3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.10.10.1 netmask 255.255.255.0 broadcast 10.10.10.255
    inet6 fe80::a46b:68ff:fe42:5b04 prefixlen 64 scopeid 0x20<link>
    ether a6:6b:68:42:5b:04 txqueuelen 1000 (Ethernet)
    RX packets 304 bytes 23921 (23.9 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 12 bytes 936 (936.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

```
root@tce-VirtualBox:~/praveen111# ip netns exec VRF1 ifconfig
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

veth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.10.10.1 netmask 255.255.255.0 broadcast 10.10.10.255
    inet6 fe80::b878:a7ff:feb0:63d7 prefixlen 64 scopeid 0x20<link>
    ether ba:78:a7:b0:63:d7 txqueuelen 1000 (Ethernet)
    RX packets 298 bytes 24829 (24.8 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 41 bytes 2322 (2.3 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

## Result:

Thus, we have performed OpenVswitch and Linux Bridge successfully.